THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL2108: PRINCIPLES OF CIVIL ENGINEERING MANAGEMENT

May 17, 2023 6:30-9:30PM

Answer ALL questions.

This is a closed-book exam.

A compound interest table is provided.

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

1. It is proposed to construct a self-financing toll highway. The criterion is that the toll paid by the vehicles over the first 30 years of the project must be sufficient to cover both the initial capital and other costs which will be incurred over the project's life.

   a) Determine a suitable charge on each vehicle passing through the highway using the following estimated figures based on 2010 prices. (15 marks)

   b) To release financial burden, the local government decides to invite a private firm to form a public-private partnership (PPP) based on Building-Own-Operate-Transfer (BOOT) scheme. Instead of the criterion listed above, the private firm aims to achieve 80% profit after 20 years' operation and then transfer the highway to the local government. Determine a suitable charge on each vehicle passing through the highway. (7 marks)

   c) Explain the concept of PPP and the benefit of adopting PPP. (8 marks)

1. Capital cost	$700,000,000
Construction cost	
Land cost	
Professional fees	
2. Annual OMR cost	$5,000,000
3. Annual administration cost	$2,500,000 for the first 30 years
4. Replacement cost of autopay ticket machine	$1,000,000 every 5 years
5. Estimated annual traffic flow	10,000 vehicles per day in the first five years; the growth rate of traffic flow is 15% every 5 years
6. Life of project	100 years
7. Required rate of return (real)	10%
8. Estimated inflation rate	10% p.a. (average)

2.

a) Describe the roles and responsibilities of a project manager throughout the building life cycle. (16 marks)

b) Discuss with examples the possible circular strategies that could be used in the construction of a new building. (9 marks)

3. The following table shows the duration and logical relationship of the activities involved in a construction project.

Activity ID	Predecessor	Duration (weeks)
A	--	5
B	A	4
C	B	8
D	A(S2, F3)	6
E	C(S4, F3), D	10
F	D(S4, F4)	8
G	F(S5, F4)	10
H	G	5
I	E, H	8

Hint: "D(S4, F4)" means activity "F" commences 4 days after the start of Activity "D" and finishes 4 days after the finish of Activity "D".

a) Prepare an activity-on-node network diagram based on the data given in the table above and identify the critical path of the project. (5 marks)

b) Prepare a table showing the early and late start, early and late finish dates of each activity. Calculate the total floats on start and on finish of Activity "C". (10 marks)

4. The university is planning to construct a new hall to accommodate more students or staff. After preliminary investigation and consulting, the university decides to adopt the Design and Build procurement methods and receives three tenders. The details of the three tenders are shown in the table below. Assuming that the rental value of the new hall is receivable at the end of each month immediately after completion. The university will make payments to the contractor at the end of every quarter. (Use the bank interest rate 1% per month.)

	Tender A	Tender B	Tender C
Construction cost	$45,000,000	$48,000,000	$60,000,000
Construction period	15 months	12 months	18 months
Rental income (per month)	$500,000	$600,000	$800,000

a) Explain the concept of Design and Build procurement method with the aid of suitable diagrams; discuss the advantage and disadvantage of this method. (10 marks)

b) Explain by calculation which tender should be selected in the time horizon of 18 months. (10 marks)

c) Assuming the university selects Tender A, if the contractor completed the contract on schedule in the first year but two months late in the last quarter (so that the payment at the end of 5th quarter was 5% of the contract sum, at the end of 16th month was 5% of the sum, at the end of 17th month was 10% of the contract sum), determine the amount that should be deducted from the final payment in order to ensure adequate compensation for the university (consider it in the time horizon of 18 months) (10 marks)

$i = 10\%$

n	caf $(1+i)^n$	pwf $\frac{1}{(1+i)^n}$	uscaf $\frac{(1+i)^n-1}{i}$	ussff $\frac{i}{(1+i)^n-1}$	uspwf $\frac{(1+i)^n-1}{i(1+i)^n}$	uscrf $\frac{i(1+i)^n}{(1+i)^n-1}$
1	1.1000	0.9091	1.0000	1.0000	0.9091	1.1000
2	1.2100	0.8264	2.1000	0.4762	1.7355	0.5762
3	1.3310	0.7513	3.3100	0.3021	2.4869	0.4021
4	1.4641	0.6830	4.6410	0.2155	3.1699	0.3155
5	1.6105	0.6209	6.1051	0.1638	3.7908	0.2638
6	1.7716	0.5645	7.7156	0.1296	4.3553	0.2296
7	1.9487	0.5132	9.4872	0.1054	4.8684	0.2054
8	2.1436	0.4665	11.436	0.0874	5.3349	0.1874
9	2.3579	0.4241	13.579	0.0736	5.7590	0.1736
10	2.5937	0.3855	15.937	0.0627	6.1446	0.1627
11	2.8531	0.3505	18.531	0.0540	6.4951	0.1540
12	3.1384	0.3186	21.384	0.0468	6.8137	0.1468
13	3.4523	0.2897	24.523	0.0408	7.1034	0.1408
14	3.7975	0.2633	27.975	0.0357	7.3667	0.1357
15	4.1772	0.2394	31.772	0.0315	7.6061	0.1315
16	4.5950	0.2176	35.950	0.0278	7.8237	0.1278
17	5.0545	0.1978	40.545	0.0247	8.0216	0.1247
18	5.5599	0.1799	45.599	0.0219	8.2014	0.1219
19	6.1159	0.1635	51.159	0.0195	8.3649	0.1195
20	6.7275	0.1486	57.275	0.0175	8.5136	0.1175
21	7.4002	0.1351	64.002	0.0156	8.6487	0.1156
22	8.1403	0.1228	71.403	0.0140	8.7715	0.1140
23	8.9543	0.1117	79.543	0.0126	8.8832	0.1126
24	9.8497	0.1015	88.497	0.0113	8.9847	0.1113
25	10.835	0.0923	98.347	0.0102	9.0770	0.1102
26	11.918	0.0839	109.18	0.0092	9.1609	0.1092
27	13.110	0.0763	121.10	0.0083	9.2372	0.1083
28	14.421	0.0693	134.21	0.0075	9.3066	0.1075
29	15.863	0.0630	148.63	0.0067	9.3696	0.1067
30	17.449	0.0573	164.49	0.0061	9.4269	0.1061

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